EXECUTIVE PRODUCT OVERVIEW

VIALACTÉE

The Autonomous, Music-Reactive LED Chandelier

100% REAL-TIME LIVE

WHAT IS VIALACTÉE?

Vialactée is a bespoke architectural chandelier composed of over 1,300 individually controlled LEDs. It continuously listens to surrounding sound and translates musical energy, rhythm, and emotion into breathing, three-dimensional light in real time. It is neither a static decorative lamp nor a simplistic sound-level flasher: it functions as an autonomous digital performer, playing live in perfect synergy with any music.

The Live Experience: Performing in Real Time

True Spontaneity — 100% Live & Unscripted

Vialactée requires zero pre-recorded stems, no MIDI files, and no pre-rendered timelines. Whether an artist plays an unreleased track, a live band improvises on stage, or guests stream a random playlist, the system ingests the live acoustic stream and synthesizes synchronized light on the fly. It reacts to the moment, completely free of manual cueing or offline preparation.

Zero Perceived Latency — The Chandelier Anticipates

Human senses immediately spot lighting delay: reacting after a beat is heard feels sluggish and clumsy. Standard sound fixtures lag because digital processing takes time. Vialactée eliminates this latency by tracking musical tempo phase and predicting upcoming beats, triggering light pulses milliseconds in advance so visual impact and speaker sound strike the audience simultaneously.

Deep Musical Intelligence vs. Volume Blinking

Instead of flashing indiscriminately whenever noise spikes, the chandelier understands musical depth. It isolates thumping sub-bass kicks, melodic mid-range chords, vocal expressions, and delicate cymbal shimmers, choreographing each frequency across distinct spatial zones of the chandelier for a nuanced, breathing performance.

Organic Narrative & Seamless Transitions

The chandelier continuously mirrors the emotional evolution of the music. A calm, intimate passage evokes gentle, breathing embers; an escalating buildup radiates energetic waves across all segments; a climatic drop unleashes explosive light cascades. Transitions between visual aesthetics morph smoothly without jarring cuts or stuttering.

Everyday Interaction: Seamless Remote Control

Instant Web Remote (Any Device)

Operates via a responsive web app reachable from any smartphone, tablet, or laptop on the local network. No installation or app store downloads required—open the browser to adjust visual modes, colors, and brightness live during any performance.

Atmosphere Curation & Presets

Explore dozens of visual modes, sculpt color palettes, and isolate individual spatial segments. Store favorite light ambiances into one-touch presets for intimate dinners, vibrant cocktail lounges, or high-energy live concerts.

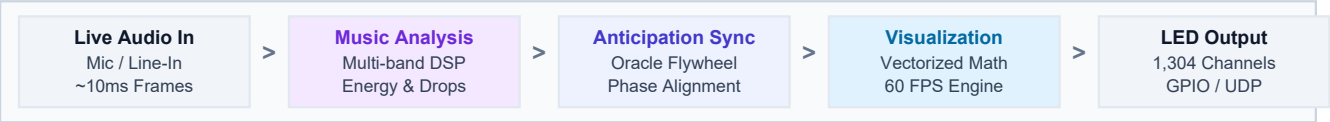
SYSTEM ARCHITECTURE & CORE ENGINEERING

TECHNICAL ARCHITECTURE

Dual Real-Time Engines: Live Music Analysis & Vectorized Visualization

100% REAL-TIME LIVE

The End-to-End Real-Time Pipeline



PART I — MUSIC ANALYSIS ENGINE

The 'Ears & Brain': Ingestion, DSP & Prediction

1. Zero-Latency Ingestion

AudioIngestion · Local_Microphone

Slices live audio into micro-windows (~10ms) from line-in or microphone feeds. Employs lock-free circular ring buffers that run independently from graphics, guaranteeing zero buffer overflows and zero latency buildup over time.

2. Multi-Band Spectral Decomposition

AudioAnalyzer · MultiBandOnset

Splits the live signal into 4 distinct frequency registers: sub-bass, low-mid, presence, and brilliance. Computes continuous spectral flux and onset novelty to track kick transients, snare hits, and melodic contours independently.

3. The 'Oracle' Beat Anticipation

comb_kernels · Phase Projection

The mathematical core of live synchronization. Rather than reacting late after a sound is emitted, the Oracle locks into the rhythmic period and projects future beat phase ahead in time, completely canceling speaker and physical travel latency.

4. Musical Context & Dynamics

MusicalContextEngine · StructuralNovelty

Extracts macro musical narrative without lookahead. Dynamically tracks cumulative energy, dynamic range, and builds/drops, signaling when the music enters an intimate breakdown or an explosive chorus drop.

PART II — VISUALIZATION ENGINE

The 'Body & Expression': Vectorized Modes & LEDs

1. Vectorized NumPy Rendering

modes.Mode · Matrix Math Pipeline

Eliminates slow, per-pixel Python loops. All light math (dimensional waves, ripples, color shifts) is computed across all 1,304 LED channels simultaneously using NumPy matrix arrays, locking silky 60 FPS output on embedded hardware.

2. 3D Spatial Geometry Mapping

Segment.py · segments_full.json

Translates abstract audio into true 3D spatial motion. Geometry profiles map physical coordinates, segment lengths, and vertical wiring directions (bottom-to-top), allowing light pulses to rise, expand, and orbit realistically in space.

3. Transition Director & Morphing

Transition_Director · Transition_Engine

Choreographs fluid transitions between visual aesthetics. Rather than jarring cuts, it continuously interpolates color palettes, speeds, and geometric states, creating an uninterrupted, evolving visual narrative.

4. Decoupled Hardware & Remote Control

HardwareFactory · Connector · wabb-interface

Decoupled drivers support dual-channel GPIO (Raspberry Pi), networked ESP32 (UDP), or a PC simulator. An asynchronous WebSocket server routes real-time commands from the React web controller without stalling animation.

Analysis Domain
4-Band Spectral DSP
~10ms Ring Buffer
Comb Beat Tracking

Visual Domain
60 FPS NumPy Math
11 Spatial Segments
Smooth Morphing

Scale & Hardware
1,304 LEDs (Full Setup)
Dual-Channel GPIO / UDP
Raspberry Pi 4 / CM4

Live Interaction
Asyncio Main Loop
WebSocket Control
React Web Interface